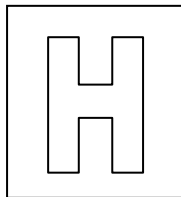


Candidate Name: _____

Class	Adm No



2021 Prelim Examination Pre-University 3

H2 Biology

9744/02

Paper 2 Structured Questions

13 September 2021

2 hours

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your Admission number and name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions in the question booklet.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question. At the end of the examination, fasten all your work securely together.

For Examiner's Use	
Section A	
1	/13
2	/10
3	/11
4	/14
5	/8
6	/11
7	/11
8	/18
9	/4
Total	/100

This question paper consists of 28 printed pages, including 2 blank pages.

[Turn over

Answer **all** questions in this section.

1. The dopamine receptor D4 (Fig. 1.1) is a G protein-coupled receptor encoded by the *DRD4* gene on chromosome 11. D4 receptor is activated by the neurotransmitter dopamine.



Fig. 1.1

(a) Describe how the primary structure of D4 gives rise to its secondary structure.

- 1) Primary structure is made up of specific number and sequence of amino acids maintained by peptide bonds;
- 2) in turn coil and fold into the specific alpha-helix maintained by hydrogen bonds;
- 3) between C=O and N-H of polypeptide backbone;

[2]

(b) Relate the structure of D4 to its role as a receptor protein.

Structure	Functions	
Consists of <u>extracellular domain</u> facing the <u>exterior</u> of the cell;	Has amino acids residue for <u>dopamine binding</u> to receive cellular <u>signals</u> ; Or <u>Complementary in shape</u> to the <u>dopamine</u> ;	
Consists of <u>intracellular domain</u> facing the <u>cytoplasm</u> of the cell;	Has available amino acids residue to <u>bind to G protein</u> to <u>relay extracellular signal</u> to intracellular molecules;	
Consists of <u>transmembrane domain</u> . <u>Amino acid</u> residues of α -helices adjacent to the cell surface membrane are <u>hydrophobic/non-polar</u> ;	<u>Hydrophobic interaction</u> between the R groups of amino acids and hydrophobic hydrocarbon/FA tail of phospholipids. This ensures GPCR is stably <u>embedded</u> in <u>cell surface membrane</u> to act as cell surface receptor;	
GPCR undergoes <u>conformational change</u> when extracellular domain binds to dopamine.	Ensures <u>intracellular domain binds/interacts with G protein</u> and results in <u>activation of G protein</u> ;	
		[4]

(c) The *DRD4* gene has a large number of alleles. Each allele has its characteristic length.

(i) Describe how gel electrophoresis reveals the differences between the different alleles of *DRD4* gene.

- 1) DNA is negatively charged due to the phosphate groups present;
 - 2) When placed in an electric field, the DNA molecules will move away from the negative electrode/cathode /towards the positive electrode/anode;
 - 3) The agarose gel impedes the movement of the DNA fragments, affecting the longer fragments more than the shorter ones/shorter fragments travel a longer distance away from the well compared to the larger fragments (OWTTE);
 - 4) Used as a molecular sieve to separate linear DNA into bands on the basis of fragment size/length/molecular mass and charge OR separate based on molecular weight
for 1 mark, max is 3 marks
 - 5) Thus, gel electrophoresis separates the different alleles of *DRD4* gene into bands, allele with longer length will be further away from the anode while allele with shorter length will be closer/nearer to the anode (OWTTE) ;
-[4]

- (ii) A scientist isolated the mRNA molecules from the cytoplasm of mouse cell and conducted gel electrophoresis to separate the mRNA molecules. Based on the number of bands which appeared on the gel, he reported that there are about 1,000 genes expressed in the cell.

Comment on the accuracy of this estimation.

- | | |
|--|-------|
| 1. The estimation is <u>not accurate</u> + at least one of the following: | |
| 2. Number of bands does not account for <u>genes expressing rRNA and tRNA</u> ; | |
| 3. mRNA of same/similar band size could have <u>different sequence</u> ; | |
| 4. <u>One gene</u> may code for more than <u>one mature mRNA</u> due to <u>alternative splicing</u> of pre-mRNA / primary RNA transcript; | |
| 5. The mRNA may not always be translated into protein, due to the presence of repressor proteins / inactivation of translation initiation factors / unstable mRNA; | |
| 6. There may be some mRNA with same sequence but different length of poly-A tail; | |
| 7. Gene are expressed in mitochondria/chloroplast; | |

.....[3]

[Total:13]

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2. Bacteria contains an enzyme that catalyse the formation of PABA.

Fig. 2.1 shows the investigation on the effect of various concentrations of inhibitor on the production of PABA.

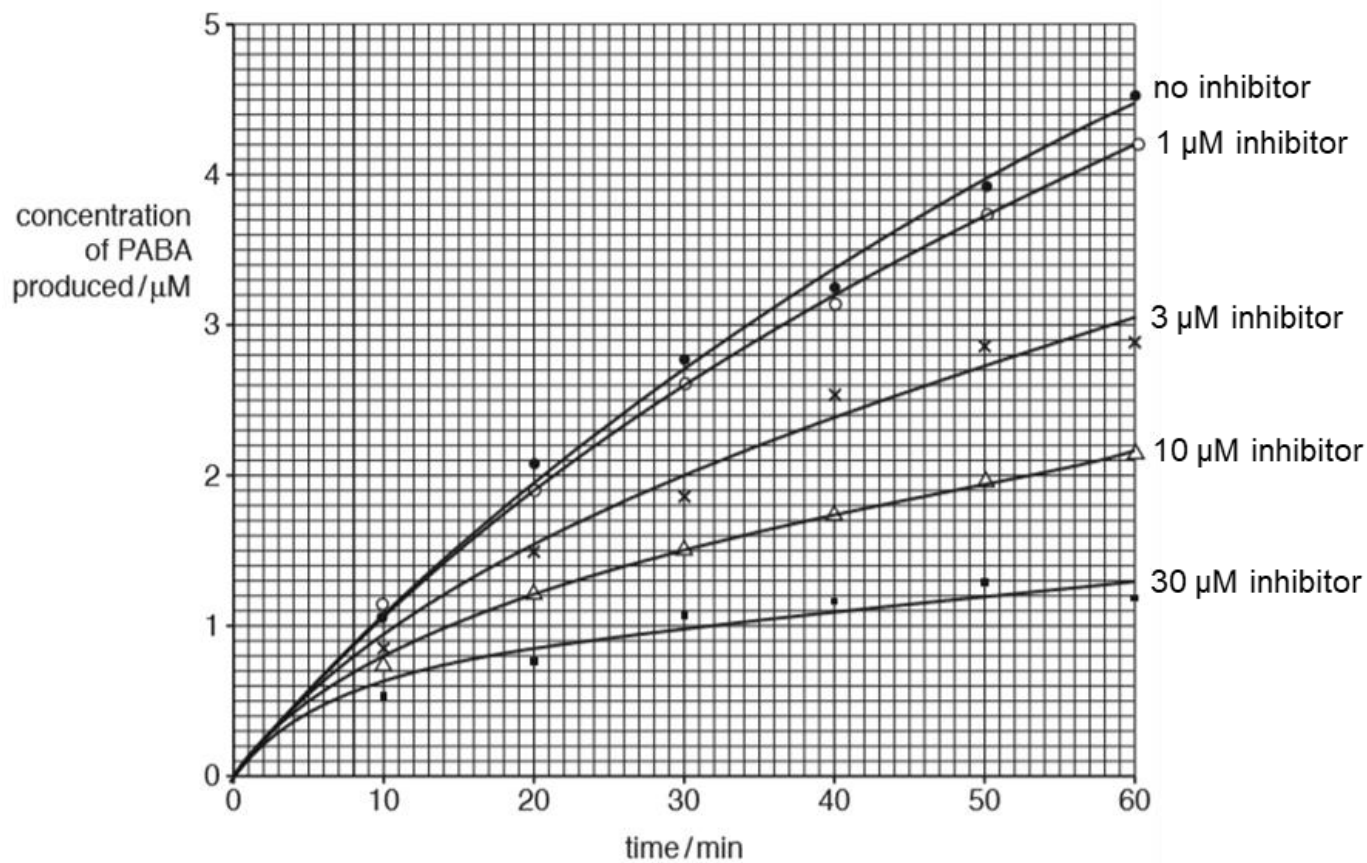


Fig. 2.1

(a) With reference to Fig. 2.1, describe the effect of inhibitor on the production of PABA.

- 1) At increasing concentration of inhibitor, concentration of PABA produced decreases as compared to no inhibitor;
- 2) From 0 min to around 2 mins, there is no difference in concentration of PABA produced in the presence of inhibitor;
- 3) As concentration of inhibitor increases from 1 μM to 30 μM , the concentration of PABA produced decreases from 42 μM to 13 μM (or any other data);
- 4) The decrease in PABA production is the least from no inhibitor to 1 μM as compared to other increase in concentration of inhibitors; OR
As concentration of inhibitor increases from 0 μM to 1 μM , the concentration of PABA produced decreases slightly from 45 μM to 42 μM ;
- 5) For all concentrations of inhibitor, rate of production becomes less steep after 5 minutes;
- 6) For last 20 minute, rate of reaction increases linearly for all concentrations of inhibitors;
AVP;

[4]

(b)

(i) Explain how a non-competitive inhibitor will affect the production of PABA.

- 1) Non-competitive inhibitor binds to a site/allosteric site other than active site of enzyme, changing the 3D conformation of the enzyme;
- 2) Active site of enzyme is no longer complementary to original substrate, thus unable bind to the original substrate;
- 3) This decrease frequency of effective collision between enzymes and substrate, decreasing the rate of formation of enzyme-substrate complex;
- 4) The amount of functional enzyme decreases;
- 5) Maximum rate of enzyme-catalysed reaction/production of PABA will be lesser even at high substrate concentration;

[3]

(ii) Explain why the results does not prove that the inhibitor works non-competitively.

- It does not show how increasing substrate concentration affects the rate of enzyme-catalysed reaction;

[1]

- (c) PABA is used to make folic acid. Folic acid is a molecule essential for the growth of most cells.

Bacteria cannot absorb folic acid from their surroundings. Folic acid from the diet is able to enter human cells but not able to cross bacterial cell walls. Human cells do not have an enzyme to make PABA.

Suggest why the inhibitor of this enzyme could be used as a drug to treat bacterial infections in humans.

-
- | |
|--|
| 1) Without the enzyme, bacteria cannot make folic acid; |
| 2) Bacteria <u>cannot grow/reproduce</u> , and will <u>die</u> ; |
-
.....

.. [2]

[Total: 10]

3. The two strands of a DNA molecule are held together by hydrogen bonds between complementary base pairs.

(a) Explain why the hydrogen bonding between the two strands of DNA is important for it to carry out its function.

- 1) Hydrogen bonds contribute to the overall 3D conformation of DNA molecule, giving it specific shape to interact with other molecules;
- 2) Many hydrogen bonds give DNA stability/strands are not easily degraded/hydrolysed/separated, allowing storage of genetic information;
- 3) Individual hydrogen bonds are easily broken down, allowing strands of DNA to be separated during DNA replication/transcription/DNA repair;
- 4) Complementary base pairing between adenine and thymine, cytosine and guanine, ensure integrity/accuracy of DNA replication/transcription/DNA repair;

[3]

(b) DNA and RNA both contain nucleotide with adenine.

Complete Table 3.1 to compare

- DNA nucleotide with adenine
- RNA nucleotide with adenine
- ATP

Table 3.1

Feature	DNA nucleotide	RNA nucleotide	ATP
Feature	DNA nucleotide with adenine	RNA nucleotide with adenine	ATP
Contains nitrogen	yes	yes	Yes;
Contains a pyrimidine base	no	no	No;
Number of phosphate groups	1	1	3;
Name of the sugar component	deoxyribose	Ribose	Ribose;

[4]

(c) Describe four differences between mRNA synthesis and polypeptide synthesis.

feature	mRNA synthesis	Polypeptide synthesis
location	nucleus	Cytoplasm;
Name of enzyme	RNA polymerase	Peptidyl transferase;
Template used by enzyme	DNA	mRNA;
Monomer used by enzyme	ribonucleotide	Amino acids;
Bonds catalysed by enzyme	Phosphodiester bond	Peptide bond;
Direction moved by enzyme with respect to template	3' to 5' of DNA template;	5' to 3' of mRNA;

1]

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4. The graph in Fig. 4.1 shows how the number of chromosomes changes during the meiotic cell cycle of an organism. The three stages of interphase are shown as G₁, S and G₂. Meiosis I and Meiosis II are shown as M_I and M_{II} respectively. M_I and M_{II} each comprise of stages A, B, C and D.

At the start of interphase, the number of chromosome per cell is 4.

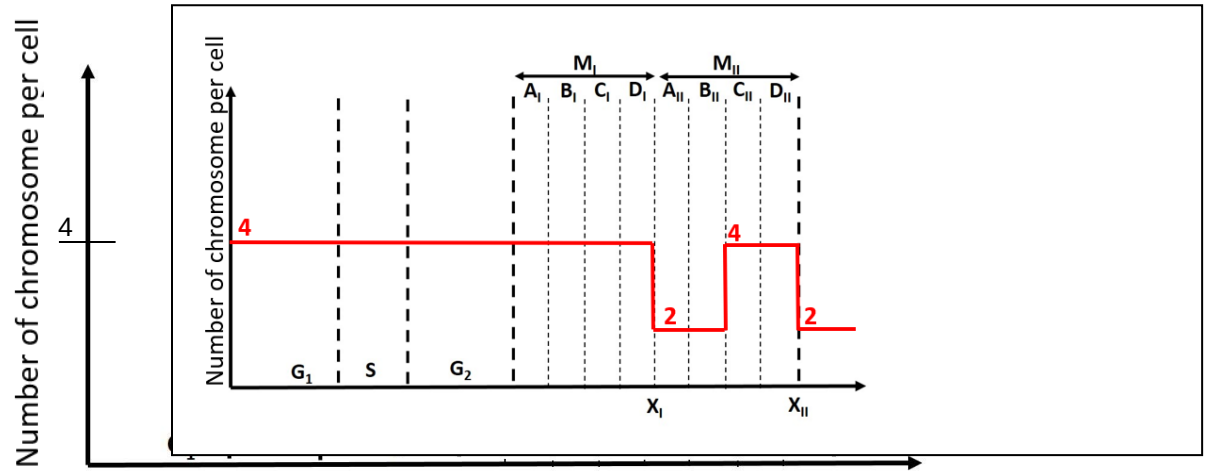


Fig. 4.1

- (a) On Fig. 4.1, draw, using line, how the number of chromosomes changes during meiosis. [2]

- Correct trend from G₁ to end of M_I;
 - Also accept if line at D_I is diagonally down.
- Correct trend from M_I to end of M_{II};
 - Also accept if line at D_{II} is diagonally down.

All correct for DNA molecules;
All correct for amount of DNA;

(b) Complete Table 4.1.

Table 4.1

Phase of Cell Cycle	Per cell	
	Number of DNA molecules	Amount of DNA(arbitrary units)
G1 phase	4	2x
G2 phase	8	4x
A _I	8	4x
D _I	8	4x
A _{II}	4	2x
D _{II}	4	2x
Cytokinesis	2	x

[2]

(c) Account for the changes in the number of chromosome and the amount of DNA per cell from G₁ till end of D_I.

- 1) During interphase, DNA undergoes replication during S phase, giving rise to a doubling of DNA content from 2x to 4x;
- 2) Although the replicated DNA now exists as two identical sister chromatids, it is still considered as a single chromosome, and thus the number of chromosome per cell remains constant at 4 throughout interphase;
- 3) The number of chromosome remains at 4 per cell from interphase till end of telophase I as anaphase I involves the separation of homologous chromosome to opposite pole of cell;
- 4) Consequently, the amount of DNA is halved from 4x to 2x and daughter cell is now haploid;

[3]

(d)

(i) Dysregulation of cell cycle can lead to cancer. In individuals with cancer, such as breast cancer, not all the cells in the patient's breast are transformed into cancer cells.

1. Development of cancer is a multi-step process that requires accumulations of gene mutations;
2. (idea of) Not all cells have accumulated sufficient mutations to be transformed into cancer cells;
3. In some cells, DNA mutation is corrected during cell cycle checkpoint/DNA repair;
Max 3 below:
4. Involves gain of function mutation of proto-oncogenes to oncogenes;
5. Loss of function mutation of tumour-suppressor genes;
6. Genes coding for telomerase is switched on/upregulated, allowing cells to divide indefinitely;
7. Genes coding for proteins in angiogenesis is up-regulated, allowing cancer cell to (idea) receive oxygen and nutrients for growth, and to remove any waste products;
8. Activation of genes that signal metastasis, resulting in malignant tumour;

[4]

(ii) Tamoxifen is a type of drugs used to treat breast cancer. It acts by blocking any estrogen present in the body from attaching to the estrogen receptor on the cancer cell. This eventually slow or prevent growth of tumours.

Suggest the normal function of estrogen in the human body.

- Estrogen stimulates cell growth;

[1]

Upon binding of estrogen to estrogen receptor, estrogen receptors move from the cytoplasm into the nucleus.

(iii) Explain why estrogen receptor moves into the nucleus.

1. Estrogen-estrogen receptor complex acts as transcription factor;
2. Binds to promoter/enhancer of proto-oncogene;

[2]

[Total: 14]

5. Coronaviruses are a group of viruses with crown-like spikes on their surface. There are four main sub-groupings of coronavirus, and one of them is the β -coronavirus. In recent studies, scientists found evidence that coronaviruses are able to hijack lysosomes and use it as an exit route from the cells.

To show that β -coronavirus do not rely on normal pathway in exiting the host cell, the scientists used Brefeldin A (BFA), an inhibitor which prevents protein transport between rough endoplasmic reticulum and golgi complex.

The results of their investigation is shown in Fig. 5.1.

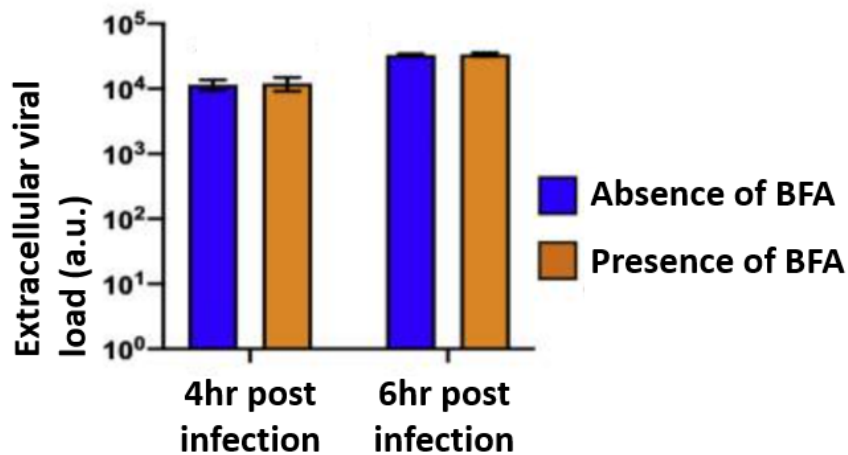


Fig. 5.1

- (a) With reference to Fig. 5.1 and the information provided, justify how the use of BFA provided evidence that β -coronavirus do not rely on the normal pathway to exit the host cell.

- 1) There is no difference in the amount of extracellular viral load detected in the presence and absence of BFA at both 4hr and 6hr post infection;
- 2) This showed that BFA does not have an effect on exit of coronavirus;

.....

.....

..... [2]

- (b)** LAMP1 is a protein that is usually found on the membrane of lysosomes. It was observed that cells infected with β -coronavirus had elevated levels of LAMP1 on the cell surface membrane.

Explain how this further supports the hypothesis that coronavirus exits the cell via lysosomal trafficking.

- 1) Elevated levels of LAMP1 on cell surface membrane is evidence that the lysosome has fused with the cell surface membrane, releasing the viral progenies out of the cell;
- 2) Upon fusion, LAMP1 is also embedded into the cell surface membrane;

[2]

- (c)** Suggest why the scientists measured the amount of virus in the extracellular environment.

- Shows that virus complete the reproductive pathway exits the host cell successfully;
- AVP;

[1]

- (d)** The team of researchers hypothesised that coronavirus were able to 'deacidify' the internal environment of lysosomes to disrupt lysosomal functions.

(i) Suggest how coronavirus might be able to 'deacidify' lysosomes.

- 1) Viral genome codes for a proton channel;
- 2) Allow for H^+ to move out of lysosome via facilitated diffusion;
OR
- 3) Viral genome codes for a protein that disrupts proton pump of lysosome;
- 4) Prevents the maintenance of high H^+ concentration in lysosome;

[2]

(ii) Suggest why it is not recommended to inhibit lysosome as part of anti-viral treatment.

- | | |
|--|---------|
| 1) Lysosome is essential in intracellular digestion/
phagocytosis/autophagy/autolysis/exocytosis; | |
| 2) Inhibiting lysosome may affect normal cellular functions; | ... [1] |

[Total: 8]

6. *Staphylococcus aureus* (*S.aureus*) is a bacteria that is found on human's skin. Methicillin resistant *Staphylococcus aureus* (MRSA) is a variety that is difficult to treat because of its resistance to methicillin and some other antibiotics.

Fig. 6.1 shows the total number of *S.aureus* infection in children under 15 years old, and the percentage of MRSA (relative to all *S.aureus* infection) in England and Wales from 1990 to 2001.

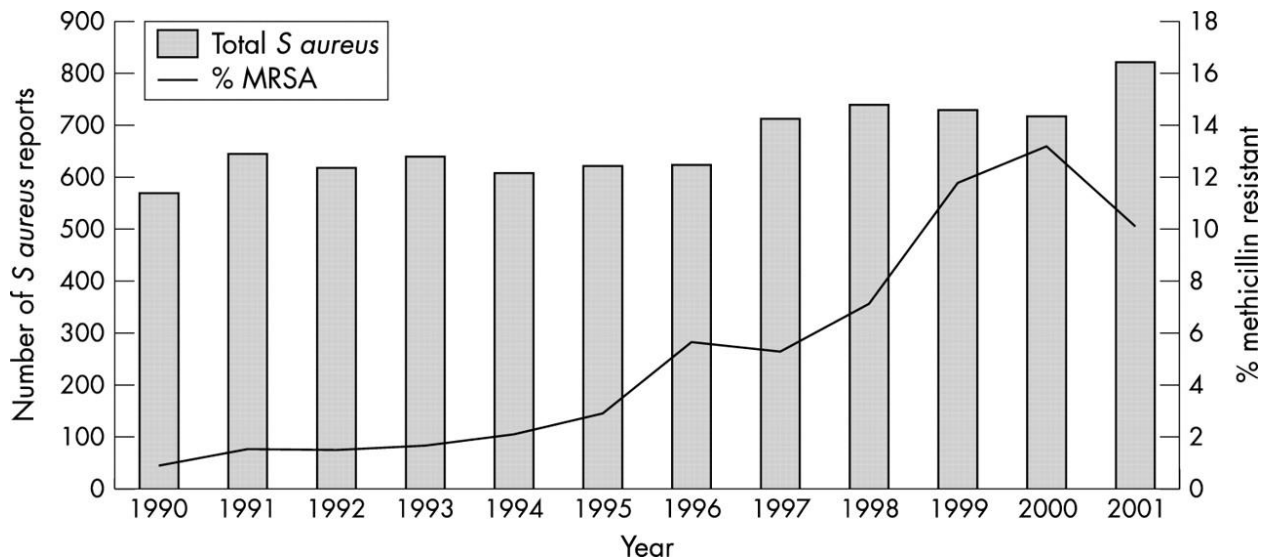


Fig. 6.1

- (a) A student claimed that the increase in percentage of MRSA is due to increase in total number of *S.aureus*.

With reference to Fig. 6.1, justify this claim.

1. The claim is not completely true/true/not true + appropriate explanation;
2. Between 1995 to 1996, the % MRSA increases from around 2% to 6%, but the total number of *S.aureus* infection remain constant at about 620 cases OR other inverse relationship;
3. However the overall trend is true as the % MRSA increases from around 1% to 10% from 1990 to 2001, the total number of *S.aureus* infection increases from about 580 cases to 800 cases;

[3]

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(b) Methicillin is a semi-synthetic penicillin.

Describe how methicillin kills bacteria.

1. Acts as competitive inhibitor and bind to active site of enzyme involved in cell wall synthesis;
2. Formation of cell wall is inhibited/weakening cell wall;
3. Due to osmotic pressure, cell wall will lyse/burst, leading to death of bacteria;

..... [2]

(c) A strain of the bacterium *Pseudomonas aeruginosa*, strain R, has a gene coding for the pump and is resistant to a group of antibiotic known as β -lactam.

The MIC is the lowest concentration of antibiotics that prevents bacteria colony from forming. Two mutant strains, strain 1 and strain 2 are derived from strain R. The minimum inhibitory concentration (MIC) of the β -lactam for strain R, mutant 1 and mutant 2 were determined. Mutant 1 and 2 differ from strain R in the expression of the gene coding for the pump.

The MICs for the three strains of *Pseudomonas aeruginosa* are shown in Table 6.1.

Table 6.1

Strain of <i>Pseudomonas aeruginosa</i>	MIC of β -lactam ($\mu\text{g cm}^{-3}$)
Resistant strain R	65
Mutant strain 1	0.5
Mutant strain 2	270

With reference to Table 6.1,

(i) explain why the MICs for mutant strain 1 and 2 differ from that for strain R.

1. Mutant strain 1 has very low resistance/less resistance while mutant strain 2 has higher resistance to β -lactam as compared to strain R;
2. In strain 1, the gene for pump could not be switched on/low expression/few pumps produced/pumps out lesser antibiotics;
3. In strain 2, the gene for pump could be expressed/high expression/high expression/more pumps produced/pumps out more antibiotics;

.....

[3]

(ii) explain how a population of strain R could be replaced by mutant strain 2.

1. Through natural selection where antibiotic acts as selection pressure;
2. Mutant 2 has selective advantage in presence of between $65 \mu\text{g cm}^{-3}$ and $270 \mu\text{g cm}^{-3}$ antibiotics;
3. Strain R bacteria are at selective disadvantage, unable to survive and die;
4. Mutant 2 is able to reproduce and pass on mutation/resistance to offspring over many generation;

.....

[3]

[Total: 11]

7. The Hawaiian archipelago is a group of volcanic islands in the Pacific Ocean. Fig. 7.1 shows the relative locations of five of these islands.



Fig. 7.1

Table 7.1 shows the size and age of these five islands and the total number of *Mecyclothorax* ground beetle species and their species density, on each island.

Data for the island of Maui is shown as two distinct regions, West and Haleakalā. This is because they formed at different times from two separate volcanoes.

Table 7.1

island	area / km ²	age of island / million years	total number of <i>Mecyclothorax</i> species	species density / number of species per km ²
Hawai'i	10 433	0.4	30	0.003
Lāna'i	364	1.3	3	0.008
Maui (West)	443	1.3	27	0.061
Maui (Haleakalā)	1440	1.1	116	0.081
Moloka'i	673	1.9 – 1.8	43	0.064
O'ahu	1545	3.7 – 2.6	20	

(a)

- (i) Complete Table 7.1 by calculating the density of *Mecyclothorax* beetle species on the island of O'ahu. [1]

0.0129/0.0131;

- (ii) Using Table 7.1, suggest why the island of Hawai'i has the lowest density of *Mecyclothorax* beetle species.

- Hawaii is formed only 0.4 million years ago, being the youngest island, thus there is less time for the beetles to arrive/colonise/speciate;
 - Hawaii has a large area of 10433 km², so density is low;

..... [2]

- (iii) Using Table 7.1 and Fig. 7.1, suggest why O'ahu has a lower number of *Mecyclothorax* beetle species than Moloka'i.

- Molokai is surrounded by three islands, as compared to Oahu which is next to only one island;
 - More beetles may reach Molokai as compared to Oahu;

..... [1]

Biological species concept is one of the species concept used to define a species.

(b) State two limitations of biological species concept.

- Cannot be applied to fossils;
 - Cannot be applied to asexual organisms as they do not interbreed;
 - There can still be gene flow between species that are morphologically distinct;
 - There can still be gene flow between species that are ecologically distinct;

..... [2]

- (c) The rate of speciation of *Mecyclothorax* beetles on the slopes of the volcano Haleakalā, on Maui, is the highest recorded for any genus or location on earth.

The volcano last erupted 400 years ago. This produced lava flows which cut through ancient forest and vegetation, dividing it into many separate microhabitats.

Explain how the large number of *Mecyclothorax* beetle species on Haleakalā developed.

- | | |
|--|------------|
| • <u>Allopatric speciation</u> arose <u>from geographical isolation/reduced gene flow</u> ; | |
| • Genetic drift/founder effect; | |
| • <u>Natural selection</u> , genetic drift or mutation occur in each sub-population due to <u>different selection pressure</u> ; | |
| • <u>Pre-existing variation</u> in each sub-population due to <u>spontaneous mutation</u> ; | |
| • Individuals with genes coding for advantageous traits are selected for and survived till reproductive age, passing down these alleles to their offsprings; | |
| • Over time, <u>allele frequencies</u> in each sub-population altered differently/genetic difference between the gene pool arise; | |
| • Each sub-population separately evolves <u>reproductive isolation</u> mechanism; | |
| | . [5] |
| | [Total:11] |

8. Fig. 8.1 shows the series of events that occur when naïve B lymphocytes encounter its specific antigen.

Upon binding to specific antigen, B lymphocyte is activated and undergoes cell growth. Activated B lymphocytes undergo mitosis to form a pool of B lymphocytes.

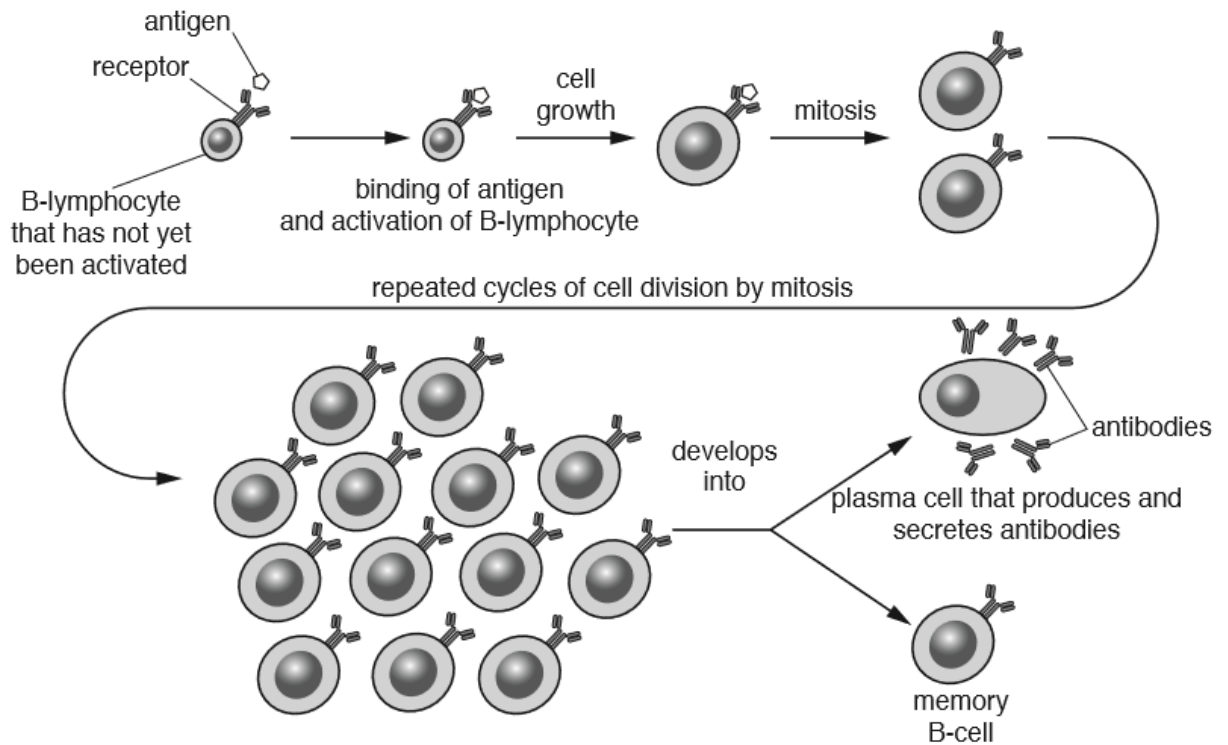


Fig. 8.1

(a) Explain the importance of forming a pool of B lymphocytes.

1. Give rise to large number of B-lymphocytes to differentiate into plasma cells and memory cells;
2. Faster production of memory and effector cells;
3. Stronger responses of adaptive immune response through actions of antibodies;
4. Ref to affinity maturation

[2]

The pool of B lymphocytes differentiate into either plasma cells or memory B cells.

(b) Explain how B lymphocytes differentiate into memory B cells, and describe the roles of memory B cells.

1. Activators, binds to enhancer of gene coding for protein essential for memory cells;
2. Repressors bind to silencer of gene coding for protein involved in plasma cells;
3. This stimulates/increase gene expression of gene essential for the memory cells and decrease gene expression (idea) of genes essential for plasma cells;
4. Cytokine released triggered differentiation

Max 2

5. Antigen receptor of memory B lymphocytes have higher affinity to the antigen;
6. Upon binding to antigen, memory B lymphocytes differentiate into plasma cells to secrete antibody
7. Stronger and faster secondary immune response;

[5]

(c) Antibodies are glycoproteins.

(i) Name and describe the structure of the organelle involved in the synthesis and chemical modification of antibodies.

Name

Structure

- Rough endoplasmic reticulum;
- Interconnecting flattened membranes/cisternae;
- Membrane of rER attached with ribosomes;

[3]

(ii) Suggest why the first antibody synthesised is not secreted.

1. They act as B cell antigen receptor on the cell surface membrane of B cell;
2. To bind to complementary antigen;

[2]

(iii) The genes responsible for antibody production are found on different chromosomes, such as chromosomes 2 and 14 in humans.

Explain how one antibody molecule is the product of more than one gene.

1. Antibody is made up of 2 heavy chains and 2 light chains;
2. (idea) Each polypeptide chain is coded by one gene;
3. Gene coding for enzyme to attach carbohydrate is coded by different gene;

[2]

(iv) Human has to ability to make millions of different antibodies. Class switching is one of the processes that give rise to the different antibody molecules.

Explain how class switching give rise to the many variety of antibodies.

1. switching its isotype classes through replacing the heavy chain constant region;
2. regulated by cytokines released by CD4+ T helper cells in the adaptive immune response;
3. Enables the same variable heavy exon to be associate with different constant heavy exon;
4. This provides antibody with the same antigen specificity but distinct effector function;
5. Increase functional diversity of antibody by replacing the constant region.

[4]

[Total:18]

9. Dengue fever is a mosquito-borne viral disease and a regular epidemic in Thailand. The peak of the dengue epidemic period is around June to August during the rainy season. It is believed that climate is an important factor for dengue transmission.

Fig. 9.1 shows how changes in temperature affects the average monthly incidence rate of dengue cases in Chiang Rai, Thailand from 2004 – 2014.

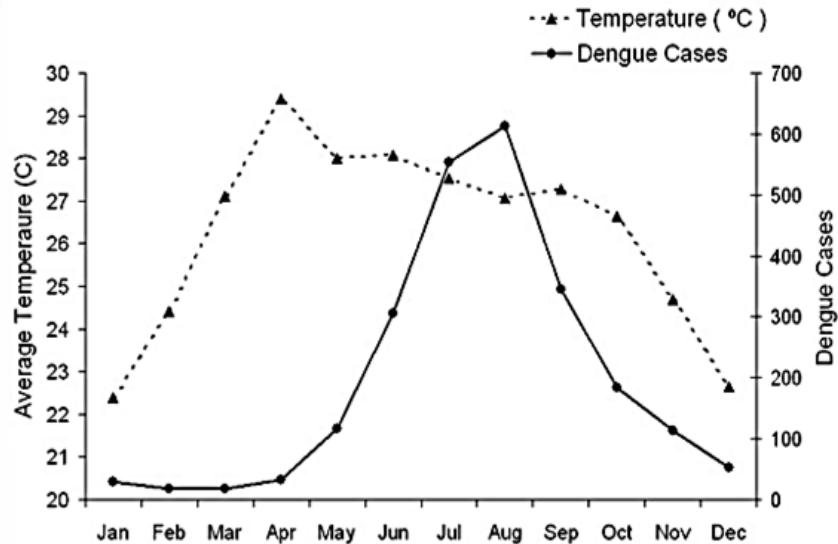


Fig. 9.1

- (a) State how each data point in Fig. 9.1 is obtained.

- By calculating the average number of dengue cases of all temperature in the same month from each year of 2004 to 2014; (idea of average of same month from every year)

[1]

- (b) With reference to Fig. 9.1, explain the trend of incidence of dengue cases with regard to changes in temperature.

- Increase in mosquitoes case is observed after increase in temperature as time is needed for mosquito to mature(idea of subsequent increase);
- Increase in temperature will lead to increase in metabolic activity of mosquitos, due to increase in frequency of effective collision between enzyme and substrate/forming more enzyme-substrate complex formed;
- Increase in maturation rate and shorter time needed to complete life cycle;
- More adult mosquito available as vector to transmit dengue virus;

[3]

[Total:4]

End of Paper